

(Please write your Enrolment No. immediately)

Enrolment No: | | | | | | | | | | | | | | | |

MID-TERM EXAMINATION
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
B.Tech — III Semester • January, 2023
Paper Code: CIC-205 • Subject: Discrete Mathematics

Time: 1½ Hours

Maximum Marks: 30

Note: Attempt **Q. No. 1** which is compulsory and **any two** more questions from the remaining.

Q. No.	Question	Max. Marks	CO(s)
Q1 (a)	Evaluate inverse, converse and contrapositive of a conditional statement $p \rightarrow q$.	[2]	CO1
Q1 (b)	Create the equivalent expression for the given expressions: (I) $\neg \exists x P(x)$ (II) $\neg \forall x P(x)$	[2]	CO1
Q1 (c)	Analyse the partially ordered set (POSET)? Give an example.	[2]	CO2
Q1 (d)	Evaluate the condition of the function to be surjective? Give an example of it.	[2]	CO2
Q1 (e)	Apply the Divide and Conquer technique on any computer problem. Give an example of it.	[2]	CO3
Q2 (a)	Show that $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ using rules of logical equivalence.	[5]	CO1
Q2 (b)	Analyse the different solution methods of homogeneous and non-homogeneous recurrence relations. Explain with an illustrative example.	[5]	CO2
Q3 (a)	Draw the following set operations with the help of Venn diagrams: (I) Union ($A \cup B$) (II) Intersection ($A \cap B$) (III) Disjoint sets ($A \cap B = \emptyset$) (IV) Set Difference ($A \setminus B$)	[5]	CO1
Q3 (b)	Let $A = \{1, 2, 3, 4, 6\}$ and R is a relation on Set A such that aRb if $a \mid b$ (a divides b). Check whether Relation R is a POSET.	[5]	CO2
Q4 (a)	In how many ways can a team of 11 cricketers be chosen from 6 bowlers, 4 wicket-keepers and 11 batsmen to give a majority of batsmen if at least 4 bowlers are to be included and there is exactly one wicket-keeper?	[5]	CO1
Q4 (b)	Minimize the given Boolean function using a 4-variable K-map: $F(A, B, C, D) = ABC'D' + ABC'D + AB'C'D + ABCD + AB'CD + ABCD' + AB'CD'$ Note: Prime notation (') denotes complementation (e.g., $A' = \neg A$). Equivalent to minterms $\Sigma m(9, 10, 11, 12, 13, 14, 15)$.	[5]	CO2